ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

MECHATRONICS

PAPER 1 · PRACTICE G

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 4. Laid out like the DHET Mechatronic Systems paper (project based, 100 marks): design specifications, automating a shelf-drilling station with pneumatic cylinders, a displacement-step diagram and a ladder diagram, mounting methods, bills of materials, critical path, Gantt chart and risk register, cold and hot commissioning, and lubrication, air leaks, MTBF, MTTR, availability and lockout.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL calculations. Draw sketches, tables, diagrams and circuits neatly. Use only a black or blue pen.

> **PROJECT.** A furniture factory drills the holes for shelf pins in wooden shelves by hand, and the holes are often in the wrong place. The factory wants an automated drilling station: the operator places a shelf against fixed stops and presses two start buttons at the same time. Pneumatic cylinder A clamps the shelf (A+), the drill motor starts and cylinder B feeds the drill down (B+), the drill returns (B−), the clamp releases (A−) and a counter adds one shelf.

QUESTION 1: DESIGN SPECIFICATION

1.1 Write a design specification for the drilling station with FIVE items. Each item must be measurable where possible (for example a number with a unit). (5)

1.2 Explain why the items of a design specification should be measurable, and give ONE example from QUESTION 1.1. (3)

1.3 Give TWO ways in which the design team can collect information from the client and the operators. (2)

[10]

QUESTION 2: AUTOMATING THE DRILLING STATION

2.1 Draw the displacement-step diagram for the sequence A+ B+ B− A−. (6)

2.2 Name FIVE pneumatic components needed for the drilling station and give the function of EACH. (5)

2.3 Draw a ladder diagram for the drilling station. Two start buttons S1 and S2 (NO) must BOTH be pressed to energise the clamp solenoid Y1 (latched). When reed switch a1 shows that the clamp is closed, the drill motor M and the feed solenoid Y2 switch on. When reed switch b1 shows full depth, a memory relay K2 latches and switches Y2 off so the drill returns. When b0 (drill up) and K2 are both on, the clamp is released and counter C1 counts one shelf. Include a STOP button. (10)

2.4 Name TWO methods of fastening the cylinders to the steel frame and give ONE advantage of EACH. (4)

[25]

QUESTION 3: PROJECT MANAGEMENT

3.1 Draw up a bill of materials for the drilling station with suitable column headings and at least FIVE items. (8)

3.2 Explain what the critical path of a project is and why the project manager must know it. (4)

3.3 Draw a Gantt chart for SIX activities of the project over a period of 8 weeks. Some activities must overlap. (8)

3.4 Explain what a risk register is. Give TWO risks for this project and how EACH can be reduced. (6)

3.5 Explain why a prototype of the station is built and tested before the final machine. (4)

[30]

QUESTION 4: TESTING AND COMMISSIONING

4.1 Explain the difference between cold commissioning and hot commissioning. (4)

4.2 Draw up a test checklist with FIVE checks that must be done before the station is handed over. (5)

4.3 During testing, the drill reaches full depth but does not return. Give THREE possible causes. (3)

4.4 Explain why the emergency stop and the two-hand start must be tested first. (3)

[15]

QUESTION 5: MAINTENANCE

5.1 Explain the purpose of a lubrication chart, and name TWO parts of the drilling station that need regular lubrication. (4)

5.2 Explain how to check the pneumatic system for air leaks. (3)

5.3 In 1 200 hours of operation the station broke down 4 times, and the repairs took 24 hours altogether. Calculate the: (6)

5.3.1 mean time between failures (MTBF) (2)

5.3.2 mean time to repair (MTTR) (2)

5.3.3 availability of the station as a percentage (2)

5.4 Describe the lockout/tagout procedure that must be followed before maintenance is done on the station. (4)

5.5 Name THREE safety signs or warning notices that must be displayed on the station. (3)

[20]

TOTAL: 100

END OF QUESTION PAPER